# Notes on Chapter 7

Throughout, $m > 1$ is the modulus and $a$ is an integer. The subject of these notes is the set of invertible elements modulo $m$, which is denoted $U_m$. The main result, Theorem 23, gives a complete classification of those $m$ for which there exists an element $\zeta \in U_m$ such that $U_m$ is equal to $\{\zeta^i : 1 \leq i \leq \phi(m)\}$. In case this is true, we say $\zeta$ is a primitive root modulo $m$. This is an important component of the complete description of the group $U_m$. For a complete description of the groups $U_m$, see [2, Remark 2.3.31].

**Definition 1.** ([1, Definition 7-1]) We say the *order of $a$ modulo $m$* is $n$ if

(1) $a^n \equiv 1 \pmod{m}$,
(2) $n > 0$, and
(3) if $t$ is a positive integer such that $a^t \equiv 1 \pmod{m}$, then $n \leq t$.

We write $|a| = n$.

**Theorem 2.** *([1, Theorem 7-1]) The order of $a$ modulo $m$ exists if and only if* $\gcd(a, m) = 1$.

*Proof.* If $\gcd(a, m) = 1$, then $a^{\phi(m)} \equiv 1 \pmod{m}$, by Euler's Theorem ([1, Theorem 5-2]). Since $\phi(m) \geq 1$, the set $\{x \geq 1 : a^x \equiv 1 \pmod{m}\}$ is nonempty. By the Well Ordering Principle, the order of $a$ exists. Conversely, suppose $|a| = n$. Then $a^n \equiv 1 \pmod{m}$, which implies

$$aa^{n-1} = 1 + mq$$

for some $q$. By Bézout's Identity, $\gcd(a, m) = 1$. □

**Theorem 3.** *([1, Theorem 7-2]) Suppose the order of $a$ modulo $m$ is $n$ and $t$ is an integer. Then $a^t \equiv 1 \pmod{m}$ if and only if $n \mid t$.*

*Proof.* Suppose $a^t \equiv 1 \pmod{m}$. Apply the Division Algorithm to write $t = nq + r$, where $0 \leq r < n$. Then

$$1 \equiv a^t \equiv (a^n)^q a^r \equiv a^r \pmod{m}.$$

By the minimal choice of $n$, we conclude that $r = 0$. Conversely, if $t = nq$, then $a^t = (a^n)^q \equiv 1 \pmod{m}$. □

**Corollary 4.** *(Corollary to [1, Theorem 7-2]) If the order of $a$ modulo $m$ is $n$, then $n \mid \phi(m)$.*

*Proof.* This follows immediately from Euler's Theorem ([1, Theorem 5-2]) and Theorem 3. □

**Corollary 5.** *(Corollary to [1, Theorem 7-2]) If the order of $a$ modulo $m$ is $n$, then $a^i \equiv a^j \pmod{m}$ if and only if $i \equiv j \pmod{n}$.*

*Proof.* If $i \equiv j \pmod{n}$, then $i = j + nq$ for some $q$. Then

$$a^i = (a^n)^q a^j \equiv a^j \pmod{m}.$$

If $a^i \equiv a^j \pmod{m}$, then $a^{i-j} \equiv 1 \pmod{m}$. By Theorem 3, $n \mid (i - j)$. Hence $i \equiv j \pmod{n}$. □

**Definition 6.** Let $\zeta \in U_m$. If the order of $\zeta$ modulo $m$ is $\phi(m)$, then we say $\zeta$ is a *primitive root modulo $m$*.

**Theorem 7.** *(*[1, Theorem 7-3]*) If $\zeta$ is a primitive root modulo $m$, then the set*

$$R = \left\{\zeta, \zeta^2, \ldots, \zeta^{\phi(m)}\right\}$$

*is a reduced residue system modulo $m$.*

*Proof.* By Corollary 5, elements of $R$ are pairwise incongruent. Hence the set $R$ has $\phi(m)$ elements. Define $f : R \to U_m$ by setting $f(\zeta^i)$ equal to the remainder upon division of $\zeta^i$ by $m$. Then $f$ is one-to-one by Corollary 5. Both sets have cardinal number $\phi(m)$, so the Pigeon Hole Principle says $f$ is onto. □

**Theorem 8.** *(*[1, Theorem 7-4]*) If the order of $a$ modulo $m$ is $n$, then the order of $a^k$ is equal to $n/\gcd(k, n)$.*

*Proof.* Set $|a^k| = \ell$. Then $a^{k\ell} \equiv 1 \pmod{m}$. Theorem 3 says $n \mid k\ell$. Write $k\ell = nx$ for some $x$. Let $d = \gcd(k, n)$ and write $k = dk_1$ and $n = dn_1$. We prove that $\ell = n_1$. By Bézout's Identity, $d = ku + nv$ for some $u$, $v$. Cancel $d$ from both sides to get $1 = k_1u + n_1v$. Therefore, $\gcd(k_1, n_1) = 1$. Cancel $d$ from $k\ell = nx$ to get $k_1\ell = n_1x$. By Euclid's Lemma, $n_1 \mid \ell$. Now look at

$$\left(a^k\right)^{n_1} \equiv a^{dk_1n_1} \equiv (a^n)^{k_1} \equiv 1 \pmod{m}.$$

Theorem 3 says $\ell$ divides $n_1$. Taken together, this shows $\ell = n_1$. □

**Corollary 9.** *(*[1, Corollary 7-1]*) If $\zeta$ is a primitive root modulo $m$, then $\zeta^i$ is a primitive root modulo $m$ if and only if $\gcd(i, \phi(m)) = 1$.*

*Proof.* By Theorem 8, the order of $\zeta^i$ is equal to $\phi(m)$ if and only if $\gcd(i, \phi(m)) = 1$. □

**Corollary 10.** *(Corollary to* [1, Theorem 7-4]*) Suppose the order of $a$ modulo $m$ is $n$. The set $\left\{|a^i| : 1 \le i \le n\right\}$ is equal to the set* $\mathrm{Div}(n)$.

*Proof.* By Theorem 8, the order of $a^i$ is equal to $n/\gcd(i, n)$, which is a divisor of $n$. If $d \mid n$, then by Theorem 8, the order of $a^{n/d}$ is equal to $d$. □

**Definition 11.** If the order of $a$ modulo $m$ is $n$, define the set

$$\langle a\rangle = \{a, a^2, \ldots, a^n\}.$$

**Corollary 12.** *(Corollary to* [1, Theorem 7-4]*) Suppose the order of $a$ modulo $m$ is $n$. For any $d$ in* $\mathrm{Div}(n)$*, there are exactly $\phi(d)$ elements of order $d$ in $\langle a\rangle$.*

*Proof.* In the notation of [1, Theorem 6-1], set

$$T_{n/d} = \{k : 1 \le k \le n \text{ and } \gcd(k, n) = n/d\}.$$

The elements of order $d$ in $\langle a\rangle$ is the set

$$\left\{a^k : 1 \le k \le n \text{ and } |a^k| = d\right\},$$

which by Theorem 8 is equal to the set

$$\left\{a^k : 1 \le k \le n \text{ and } \gcd(k, n) = n/d\right\},$$

which is equal to

$$\left\{a^k : 1 \le k \le n \text{ and } k \in T_{n/d}\right\}.$$

The cardinality of the last set is $\#\left(T_{n/d}\right)$, which by [1, Theorem 6-1] is equal to $\phi(d)$. □

**Theorem 13.** *([1, Theorem 7-5]) If there exists a primitive root modulo $m$, then there are exactly $\phi(\phi(m))$ mutually incongruent primitive roots modulo $m$.*

*Proof.* Apply Corollary 12 with $d = n = \phi(m)$. □

**Corollary 14.** *(Corollary to [1, Theorem 7-4]) If $a_1$ and $a_2$ are invertible modulo $m$, the order of $a_1$ modulo $m$ is $d_1$, the order of $a_2$ modulo $m$ is $d_2$, and $\gcd(d_1, d_2) = 1$, then the following are true.*

*(1) $\langle a_1 \rangle \cap \langle a_2 \rangle = \{1\}$. That is, $a_1^i \equiv a_2^j \pmod{m}$ if and only if $a_1^i \equiv 1 \pmod{m}$ and $a_2^j \equiv 1 \pmod{m}$.*
*(2) The order of $a_1 a_2$ modulo $m$ is $d_1 d_2$.*

*Proof.* (1): Suppose $a_1^i \equiv a_2^j \pmod{m}$ for some $i$ and $j$. Let $d = |a_1^i| = |a_1^i|$. Theorem 8 says $d \mid d_1$ and $d \mid d_2$. Therefore, $d \leq \gcd(d_1, d_2) = 1$. Hence $a_1^i \equiv 1 \pmod{m}$ and $a_2^j \equiv 1 \pmod{m}$. The converse is immediate by [1, Theorem 4-1].

(2): By [1, Exercise 4 of §2.2], $\operatorname{lcm}(d_1, d_2) = d_1 d_2$. Let $|a_1 a_2| = d$. Then

$$(a_1 a_2)^{d_1 d_2} = \left(a_1^{d_1}\right)^{d_2} \left(a_2^{d_2}\right)^{d_1} \equiv 1 \pmod{m}.$$

Theorem 3 says $d \mid d_1 d_2$. Since $(a_1 a_2)^d \equiv 1 \pmod{m}$, we have $a_1^d \equiv a_2^{-d} \pmod{m}$. Part (1) together with Theorem 3 imply that $d_1 \mid d$ and $d_2 \mid d$. Hence, $d_1 d_2 = \operatorname{lcm}(d_1, d_2) \leq d$. Taken together we have $d_1 d_2 = d$. □

**Corollary 15.** *(Corollary to [1, Theorem 7-4]) If $m > 2$ and $n > 2$ and $\gcd(m, n) = 1$, then there is no primitive root modulo $mn$.*

*Proof.* By [1, Exercise 4 of § 6-1], $\phi(m)$ is even and $\phi(n)$ is even. Because $m > 2$, $1 \not\equiv -1 \pmod{m}$. There are at least two solutions modulo $m$ to $x^2 \equiv 1 \pmod{m}$, namely $x \equiv \pm 1 \pmod{m}$. Likewise, there are at least two solutions modulo $n$ to $x^2 \equiv 1 \pmod{n}$, namely $x \equiv \pm 1 \pmod{n}$. By the Chinese Remainder Theorem, there is a unique $x_1$ in $U_{mn}$ which satisfies

$$\begin{cases} x_1 \equiv -1 \pmod{m} \\ x_1 \equiv 1 \pmod{n} \end{cases}$$

Likewise, there exist unique $x_2, x_3, x_4$ in $U_{mn}$ which satisfy

$$\begin{cases} x_2 \equiv 1 \pmod{m} \\ x_2 \equiv -1 \pmod{n} \end{cases}, \begin{cases} x_3 \equiv -1 \pmod{m} \\ x_3 \equiv -1 \pmod{n} \end{cases}, \begin{cases} x_3 \equiv 1 \pmod{m} \\ x_3 \equiv 1 \pmod{n} \end{cases}.$$

Hence $x_1, x_2, x_3, x_4$ are four mutually incongruent solutions to $x^2 \equiv 1 \pmod{mn}$. For contradiction's sake, assume $\zeta$ is a primitive root modulo $mn$. By Corollary 12, there are only two solutions to $x^2 \equiv 1 \pmod{mn}$, namely $x \equiv \pm 1 \pmod{mn}$. This is a contradiction, which implies a primitive root does not exist. □

**Theorem 16.** *For all $k \geq 3$ the following are true.*

*(1) For any odd integer $a$, $a^{2^{k-2}} \equiv 1 \pmod{2^k}$.*
*(2) The order of $a$ modulo $2^k$ is less than or equal to $a^{2^{k-2}}$.*
*(3) There is no primitive root modulo $2^k$.*

*Proof.* (1): The proof is by induction on $k$. For $k = 3$, this follows from direct computation: $1^2 \equiv 3^2 \equiv 5^2 \equiv 7^2 \equiv 1 \pmod{8}$. Inductively assume $k \geq 3$, $a$ is odd,

and $a^{2^{k-2}} \equiv 1 \pmod{2^k}$. Then $a^{2^{k-2}} = 1 + x2^k$ for some integer $x$. Squaring both sides leads to the following.

$$\begin{aligned} a^{2^{k-1}} &= 1 + 2x2^k + x^2 2^{2k} \\ &= 1 + \left(x + x^2 2^{k-1}\right) 2^{k+1} \\ &\equiv 1 \pmod{2^{k+1}}. \end{aligned}$$

This and Mathematical Induction prove (1).

(2): This is straight from Part (1) and Definition 1.

(3): The set of units modulo $2^k$ is equal to $U_{2^k} = \{u : 1 \leq u \leq 2^k \text{ and } u \text{ is odd}\}$. By [1, Example 6-1], $\phi(2^k) = 2^{k-1}$. By (2) the order of every unit in $U_{2^k}$ is less than $\phi(2^k)$. □

**Theorem 17.** *(*[1, Theorem 7-6]*) If $p$ is a prime, then there exists a primitive root modulo $p$.*

*Proof.* If $p = 2$, then $\phi(2) = 1$ and 1 is a primitive root. From now on assume $p$ is an odd prime. Let $u \in U_p$ and assume the order of $u$ is $n$ modulo $p$. By Corollary 5, the set $\langle u \rangle = \{u, u^2, \ldots, u^n\}$ is a set of $n$ numbers pairwise incongruent modulo $p$. For any integer $i$, the order of $u^i$ divides $n$, by Theorem 8. By Theorem 3, every $u^i$ is a solution to the polynomial congruence

$$x^n \equiv 1 \pmod{p}. \tag{1}$$

So $\{u, u^2, \ldots, u^n\}$ is a set of $n$ mutually incongruent solutions to Eq. (1). By [1, Theorem 5-5], the polynomial congruence Eq. (1) has no more than $n$ such solutions. Thus every solution to Eq. (1) is congruent to an element of $\{u, u^2, \ldots, u^n\}$. In particular, every element of order $n$ in $U_p$ is congruent to an element of $\{u, u^2, \ldots, u^n\}$. In Corollary 12 we proved that the number of elements in $\{u, u^2, \ldots, u^n\}$ of order $n$ is equal to $\phi(n)$. Define an arithmetic function $N : \{1, 2, 3, \ldots\} \to \mathbb{N}$ by setting $N(n) = \#\{a \in U_p : |a| = n\}$. Thus $N(n)$ is the number of elements in $U_p$ of order $n$. We have shown that if $N(n) \neq 0$, then $N(n) = \phi(n)$. By Corollary 4, if $a \in U_p$, then $|a|$, the order of $a$, is a divisor of $\phi(p) = p - 1$. In particular, we have $N(n) \leq \phi(n)$, for all $n \in \mathrm{Div}(p-1)$. We have

$$p - 1 = \sum_{n|(p-1)} N(n) \leq \sum_{n|(p-1)} \phi(n) = p - 1$$

where the right-most equation is by [1, Theorem 6-1]. Subtracting

$$0 = \sum_{n|(p-1)} (\phi(n) - N(n))$$

where the right hand side is a sum of non-negative integers. We conclude that for every $n \in \mathrm{Div}(p-1)$ there are exactly $N(n) = \phi(n)$ elements of order $n$ in $U_p$. In particular, there are $\phi(p-1)$ primitive roots modulo $p$. □

**Lemma 18.** *If $p$ is an odd prime, then there exists a primitive root $\zeta$ modulo $p$ such that $\zeta^{p-1} \not\equiv 1 \pmod{p^2}$.*

*Proof.* Let $\zeta$ be an integer such that the order of $\zeta$ is $p - 1$ modulo $p$. Assume $\zeta^{p-1} \equiv 1 \pmod{p^2}$. Note that $\zeta + p \equiv \zeta \pmod{p^2}$, hence $\zeta + p$ is another primitive

root modulo $p$. We show that $\zeta + p$ satisfies the conclusion of the lemma. From the Binomial Theorem we have

$$(\zeta + p)^{p-1} = \zeta^{p-1} + (p-1)\zeta^{p-2}p + \text{terms with a factor of } p^2.$$

Reducing modulo $p^2$, we get

$$\begin{aligned}(\zeta + p)^{p-1} &\equiv \zeta^{p-1} + (p-1)\zeta^{p-2}p \pmod{p^2}\\ &\equiv 1 + \zeta^{p-2}p^2 - \zeta^{p-2}p \pmod{p^2}\\ &\equiv 1 - \zeta^{p-2}p \pmod{p^2}.\end{aligned}$$

Now $(\zeta + p)^{p-1} \not\equiv 1 \pmod{p^2}$ because $\gcd(\zeta, p) = 1$. □

**Lemma 19.** *If $0 \le k \le n$, then $\binom{n}{k}$ is a positive integer.*

*Proof.* This is a proof by Mathematical Induction on the variable $n$. We proved that $\binom{n}{k} = 1$ if $k = n$ or $k = 0$. In other words, on the boundaries of Pascal's Triangle, the numbers are all equal to 1. This serves as a basis for the induction. Inductively assume $n \ge 1$ and that the numbers in row $n$ of Pascal's triangle are positive integers. In other words, assume $\binom{n}{i}$ is a positive integer for all $i$ such that $0 \le i \le n$. By Pascal's Identity, for any $k$ such that $1 \le k \le n$, $\binom{n+1}{k} = \binom{n}{k-1} + \binom{n}{k}$, which is a positive integer. Therefore, every number in row $n+1$ of Pascal's Triangle is a positive integer. □

**Lemma 20.** *If $p$ is prime and $1 \le k \le p-1$, then $\binom{p}{k} \equiv 0 \pmod{p}$.*

*Proof.* By definition, $\binom{p}{k} = \frac{p!}{k!(p-k)!} = \frac{p(p-1)!}{k!(p-k)!}$. We know from Lemma 19 that $\binom{p}{k}$ is an integer. That is, the denominator divides the numerator. The denominator, $k!(p-k)!$, is relatively prime to $p$, since $p$ is prime. By Euclid's Lemma, the denominator divides $(p-1)!$. In other words, $\frac{(p-1)!}{k!(p-k)!}$ is an integer. □

**Lemma 21.** *If $p$ is an odd prime and $\zeta$ is a primitive root modulo $p$ such that $\zeta^{p-1} \not\equiv 1 \pmod{p^2}$, then for each $k \ge 2$,*

$$\left(\zeta^{p-1}\right)^{p^{k-2}} \not\equiv 1 \pmod{p^k}.$$

*Proof.* The proof is by Mathematical Induction, on the variable $k$. The basis step, $k = 2$, is true by hypothesis. Fix $k \ge 2$ and assume $\left(\zeta^{p-1}\right)^{p^{k-2}} \not\equiv 1 \pmod{p^k}$. Note that $1 = \gcd(\zeta, p) = \cdots = \gcd(\zeta, p^{k-1}) = \gcd(\zeta, p^k)$. By [1, Example 6-1], we have $\phi(p^{k-1}) = p^{k-2}(p-1)$, and Euler's Theorem ([1, Theorem 5-2]) says

$$\left(\zeta^{p-1}\right)^{p^{k-2}} \equiv 1 \pmod{p^{k-1}}.$$

For some integer $x$ we have

$$\left(\zeta^{p-1}\right)^{p^{k-2}} = 1 + xp^{k-1}.$$

By our induction hypothesis we know $p$ does not divide $x$. Raise both sides to the power $p$. Expand the right hand side using the Binomial Theorem to get

$$\begin{aligned}\left(\zeta^{p-1}\right)^{p^{k-1}} &= \left(1 + xp^{k-1}\right)^p\\ &= 1 + pxp^{k-1} + \sum_{i=2}^{p-1}\binom{p}{i}x^i p^{i(k-1)} + x^p p^{p(k-1)}.\end{aligned} \tag{2}$$

Since $k-2 \geq 0$, we see that $k+k-2 \geq k$, or $2(k-1) \geq k$. More generally, for any $i \geq 2$, it is true that $i(k-1) \geq k$. Applying this and Lemma 20, for $2 \leq i < p$ we see that

$$\binom{p}{i} x^i p^{i(k-1)} = \frac{p(p-1)!}{i!(p-i)!} p^{i(k-1)} x^i = \frac{(p-1)!}{i!(p-i)!} p^{i(k-1)+1} x^i$$

is a multiple of $p^{k+1}$. For $i = p$, it follows from $2(k-1) \geq k$ that $p(k-1) \geq (p/2)k$. But $p/2 \geq 3/2$, so $p(k-1) \geq (3/2)k = k + k/2$. Also, $k \geq 2$, so $p(k-1) \geq k+1$. This shows the term $p^{p(k-1)}x^p$ is a multiple of $p^{k+1}$. Reducing Eq. (2) modulo $p^{k+1}$, we have

$$\left(\zeta^{p-1}\right)^{p^{k-1}} \equiv 1 + xp^k \pmod{p^{k+1}}.$$

Since $p$ does not divide $x$, this proves $\left(\zeta^{p-1}\right)^{p^{k-1}} \not\equiv 1 \pmod{p^{k+1}}$. □

**Theorem 22.** *If $p$ is an odd prime and $a \geq 1$, then the following are true.*

*(1) There exists a primitive root modulo $p^a$.*
*(2) There exists a primitive root modulo $2p^a$.*

*Proof.* (1): By Lemmas 18 and 21, there exists a primitive root $\zeta$ for $p$ such that $\left(\zeta^{p-1}\right)^{p^{k-2}} \not\equiv 1 \pmod{p^k}$. Denote the order of $\zeta$ modulo $p^k$ by $n$. Then $n$ divides $\phi(p^k) = p^{k-1}(p-1)$. Since $\zeta^n \equiv 1 \pmod{p^k}$, this implies $\zeta^n \equiv 1 \pmod{p}$. But $\zeta$ is a primitive root for $p$, so $p-1$ divides $n$. Write $n = (p-1)p^\ell$ for some $\ell$ such that $0 \leq \ell \leq k-1$. For sake of contradiction assume $\ell \leq k-2$. Then $n$ divides $p^{k-2}(p-1)$. By Theorem 3

$$\left(\zeta^{p-1}\right)^{p^{k-2}} = \zeta^{p^{k-2}(p-1)} \equiv 1 \pmod{p^k}.$$

This contradiction implies $\ell = k-1$ and $\zeta$ is a primitive root for $p^k$.

(2): By [1, Theorem 6-4], $\phi$ is multiplicative. Hence $\phi(2p^a) = \phi(2)\phi(p^a) = \phi(p^a)$. Let $\zeta$ be a primitive root for $p^a$, which exists by Part (1). By the Chinese Remainder Theorem there exists $r$ in $U_{2p^a}$ such that

$$\begin{cases} r & \equiv \zeta \pmod{p^a} \\ r & \equiv 1 \pmod{2}. \end{cases}$$

Let $d$ denote the order of $r$ modulo $2p^a$. By Corollary 4, $d$ divides $\phi(2p^a) = \phi(p^a)$. Since $r^d \equiv 1 \pmod{2p^a}$, it follows that $r^d \equiv 1 \pmod{p^a}$. Since $r \equiv \zeta \pmod{p^a}$ and the order of $\zeta$ is $\phi(p^a)$, Theorem 3 implies $\phi(p^a)$ divides $d$. Taken together, this proves $d = \phi(p^a) = \phi(2p^a)$. □

**Theorem 23.** *A primitive root exists modulo $m$ if and only if $m$ is in one of the following forms: $m = 2$, $m = 4$, $m = p^a$, or $m = 2p^a$, where $p$ is an odd prime and $a \geq 1$.*

*Proof.* A primitive root for $m = 2$ is 1. A primitive root for $m = 4$ is $-1$. If $m = p^a$ or $m = 2p^a$, then by Theorem 22 a primitive root exists modulo $m$. Now suppose $m$ is not one of the forms listed. If $m$ is a prime power, then $m = 2^k$, where $k \geq 3$. By Theorem 16, there is no primitive root modulo $2^k$. Suppose there are two distinct odd primes, say $p$ and $q$, that divide $m$. Then $m = p^e x$, where $q \mid x$ and $\gcd(p, x) = 1$. By Corollary 15, there is no primitive root modulo $m = p^e x$. The last case is $m = 2^k x$ where $k > 1$, $x$ is odd, and $x > 1$. Again, by Corollary 15, there is no primitive root modulo $m = 2^k x$. □

## References

[1] George E. Andrews, *Number theory*, Dover Publications, Inc., New York, 1994, Corrected reprint of the 1971 original [Dover, New York; MR0309838 (46 #8943)]. MR 1298627

[2] Timothy J. Ford, *Introduction to abstract algebra*, Open Math Notes, American Mathematical Society, Reference # OMN:202006.110827, 2024, available at the URL: `https://www.ams.org/open-math-notes/omn-view-listing?listingId=110827`.